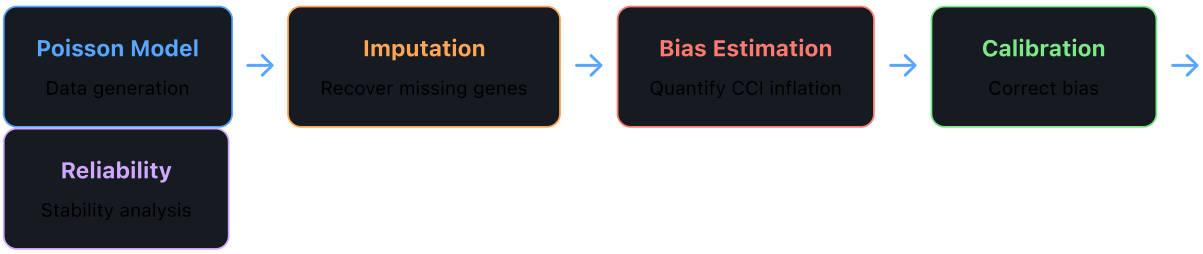
Biostatistics Formulations

Measurement Error Framework for Spatial Omics Inference — Complete Mathematical Reference

Framework Overview



Core thesis: Spatial imputation methods introduce spatially correlated reconstruction errors that inflate cell–cell interaction statistics, generating false biological interactions. This framework detects, corrects, and quantifies this artifact.

1. Poisson Measurement Model

SINGLE-CELL OBSERVATION MODEL

For cell i at spatial location $\mathbf{s}_i \in \mathbb{R}^2$ and gene g:

$$y_{ig} \sim \text{Poisson}(\delta_g \cdot \lambda_{ig}), \quad g \in S \subset \{1, \ldots, G\}$$

Symbol	Meaning	Typical Value
y_{ig}	Observed transcript count	$\{0, 1, 2, \ldots\}$
λ_{ig}	True expression rate	> 0
δ_g	Detection efficiency (probe-specific)	$(0, 1]$
S	Measured gene panel	$\lvert S\rvert = m \ll G$
G	Total genes in genome	$\approx 20{,}000$

Key Properties

MEAN-VARIANCE RELATIONSHIP

$$\mathbb{E}[y_{ig}] = \mathrm{Var}(y_{ig}) = \delta_g \lambda_{ig}$$

Signal-to-noise ratio:

$$\mathrm{SNR}_{ig} = \frac{(\delta_g \lambda_{ig})^2}{\delta_g \lambda_{ig}} = \delta_g \lambda_{ig}$$

Biological implication: SNR scales linearly with expression level. Low-expression genes (e.g., transcription factors, rare cytokines) have inherently worse SNR — this is physics, not a platform defect.

The Selection Matrix

PANEL OBSERVATION OPERATOR

$$\mathbf{y}_{i,\mathrm{obs}} = A_S \, \mathbf{y}_i \in \mathbb{Z}_+^m$$

where $A_S \in \{0,1\}^{m \times G}$ selects rows of the identity matrix corresponding to panel S:

$$(A_S)_{jg} = \begin{cases} 1 & \text{if gene } g \text{ is the } j\text{-th gene in panel } S \\ 0 & \text{otherwise} \end{cases}$$

Observation fraction: $\rho = m/G$. For Xenium 313-gene panel: $\rho \approx 0.016$ (1.6%).

2. Low-Rank Structure

GENE PROGRAM DECOMPOSITION

$$\log \lambda_{ig} = \mu_g + \sum_{k=1}^{K} u_{ik} \, v_{gk} + \epsilon_{ig}$$

In matrix form:

$$\log \mathbf{\Lambda} = \mathbf{1}\boldsymbol{\mu}^\top + \mathbf{U}\mathbf{V}^\top + \mathbf{E}$$

Symbol	Dimension	Meaning
$\boldsymbol{\mu}$	$G \times 1$	Baseline log-expression per gene

$\mathbf{U}$	$N \times K$	Cell loading matrix (program activities)
$\mathbf{V}$	$G \times K$	Gene program matrix (gene membership)
K	scalar	Number of gene programs ($K \ll \min(N, G)$)
$\mathbf{E}$	$N \times G$	Residual noise

Why Recovery Is Possible

DEGREES OF FREEDOM ARGUMENT

Without structure: $N \times G$ unknowns, $N \times m$ observations. Ratio $= G/m \approx 64\times$ underdetermined.
With rank K: Parameters $= G + K(N + G)$. Recovery requires:

$$m > \frac{G}{N} + K\left(1 + \frac{G}{N}\right)$$

For $N = 100{,}000$, $G = 20{,}000$, $K = 30$:

$$m > 0.2 + 36.0 = 36.2$$

Key insight: You need roughly $m > K$ measured genes. With ~30 gene programs, even 50 genes is theoretically sufficient. This is why imputation works at all.

3. Recovery Guarantees

POISSON MATRIX COMPLETION BOUND (CAO & XIE, 2015)

Under rank-K structure, incoherence, and uniform panel selection:

$$\boxed{\frac{1}{NG}\|\hat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_F^2 \leq C \cdot \frac{K(N+G)\,\lambda_{\max}}{Nm}}$$

For $N \gg G$, simplifies to:

$$\text{MSE} \approx C \cdot \frac{K\,\lambda_{\max}}{m}$$

Term	Effect	Meaning
K ↑	Error ↑	More gene programs = harder recovery
$\lambda_{\max}$ ↑	Error ↑	Poisson variance scales with rate
m ↑	Error ↓	Larger panel = better recovery
N ↑	Error ↓	More cells help learn gene programs

Critical ratio: K/m — rank divided by panel size governs reconstruction quality. If $K/m \to 1$, recovery degrades catastrophically.

Penalized Maximum Likelihood Estimator

NUCLEAR NORM REGULARIZED ESTIMATION

$$\hat{\mathbf{\Lambda}} = \arg\min_{\mathbf{\Lambda} \succ 0} \left[-\ell(\mathbf{\Lambda}) + \tau \|\mathbf{\Lambda}\|_*\right]$$

Log-likelihood:

$$\ell(\mathbf{\Lambda}) = \sum_{i=1}^{N} \sum_{g \in S} \left[y_{ig} \log \lambda_{ig} - \lambda_{ig} - \log(y_{ig}!)\right]$$

$\|\mathbf{\Lambda}\|_* = \sum_k \sigma_k(\mathbf{\Lambda})$ is the nuclear norm (convex surrogate for rank).

4. Per-Gene Reconstruction Error

GENE-SPECIFIC MSE DECOMPOSITION

$$\mathrm{MSE}(g) = \underbrace{\frac{1}{\delta_g \bar{\lambda}_g}}_{\text{Poisson shot noise}} + \underbrace{\|\mathbf{v}_g\|^2 (1 - R_g^2)}_{\text{Panel incompleteness}}$$

Gene Predictability (R_g^2)

LATENT-SPACE COEFFICIENT OF DETERMINATION

$$R_g^2 = \frac{\mathbf{v}_g^\top \mathbf{V}_S^\top (\mathbf{V}_S \mathbf{V}_S^\top)^{-1} \mathbf{V}_S \mathbf{v}_g}{\|\mathbf{v}_g\|^2}$$

Residual variance:

$$\sigma_g^2 = \|\mathbf{v}_g\|^2 (1 - R_g^2)$$

R_g^2 Value	Meaning	Recovery Quality
≈ 1	Gene lies in span of measured genes	Reliable
$0.5 - 0.8$	Partially predictable	Moderate
≈ 0	Orthogonal to measured panel	Unrecoverable

5. Moran's I (Global Spatial Autocorrelation)

Spatial Statistic Used in Regression Calibration

Question

"Are nearby cells more similar in expression than expected by chance?"

MORAN'S I

$$I = \frac{N}{W} \cdot \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$$

Symbol	Meaning
N	Number of cells
x_i	Expression of the gene in cell i
$\bar{x}$	Mean expression across all cells
w_{ij}	Spatial weight: 1 if i, j are neighbors, 0 otherwise

$W = \sum_i \sum_j w_{ij}$	Total weight (2 × number of edges)

Worked Example

9 cells in a 3×3 grid with gene X expression:

8	7	9
7	8	2
1	2	1

STEP-BY-STEP COMPUTATION

Step 1: Mean

$$\bar{x} = \frac{8+7+9+7+8+2+1+2+1}{9} = 5.0$$

Step 2: Deviations

```
+3    +2    +4
+2    +3    −3
−4    −3    −4
```

Step 3: Neighbor Products

Pair	$(x_i - \bar{x})$	$(x_j - \bar{x})$	Product
(1,2)	+3	+2	+6
(1,4)	+3	+2	+6
(2,3)	+2	+4	+8
(2,5)	+2	+3	+6
(3,6)	+4	-3	-12
(4,5)	+2	+3	+6
(4,7)	+2	-4	-8

(5,6)	+3	-3	-9
(5,8)	+3	-3	-9
(6,9)	-3	-4	+12
(7,8)	-4	-3	+12
(8,9)	-3	-4	+12

Step 4: Compute

$$\sum\sum w_{ij}(x_i - \bar{x})(x_j - \bar{x}) = 2 \times 30 = 60$$

$$\sum (x_i - \bar{x})^2 = 92, \quad W = 24$$

$$I = \frac{9}{24} \times \frac{60}{92} = 0.375 \times 0.652 = \mathbf{0.245}$$

Interpretation Scale

Value	Pattern	Visual
$I \approx +1$	Strong clustering (similar neighbors)	Clustered
$I \approx 0$	Random spatial distribution	Random
$I \approx -1$	Checkerboard (dissimilar neighbors)	Dispersed

Role in your framework: Moran's I of imputed gene expression estimates the *spatial correlation of imputation errors*. Graph-based methods (GNN, diffusion) produce high Moran's I in their errors because they share information across spatial neighbors. This correlated error is what inflates CCI statistics.

6. Geary's C (Local Dissimilarity)

Spatial Statistic | Detects Local Heterogeneity

Question

"How different are neighboring cells from each other?"

GEARY'S C

$$C = \frac{(N-1)}{2W} \cdot \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} w_{ij}(x_i - x_j)^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

Key Difference from Moran's I

Property	Moran's I	Geary's C
Numerator	$(x_i - \bar{x})(x_j - \bar{x})$ Deviations from *global* mean	$(x_i - x_j)^2$ *Direct* pairwise differences
Sensitive to	Global pattern	Local differences
High clustering	$I \to +1$	$C \to 0$
Random	$I \to 0$	$C \to 1$
Checkerboard	$I \to -1$	$C \to +2$

Worked Example (Same Grid)

COMPUTATION

Pair	x_i	x_j	$(x_i - x_j)^2$
(1,2)	8	7	1
(2,3)	7	9	4
(3,6)	9	2	49
(4,7)	7	1	36
(5,6)	8	2	36
(5,8)	8	2	36
(6,9)	2	1	1
(7,8)	1	2	1

Boundary pairs (3,6), (4,7), (5,6), (5,8) dominate — they are where the spatial pattern **breaks**.

$$C = \frac{8}{48} \times \frac{336}{92} = 0.167 \times 3.652 = \mathbf{0.609}$$

$C < 1$ confirms positive spatial autocorrelation (consistent with $I > 0$).

Role in your framework: Geary's C detects *local* heterogeneity in imputation quality. If errors are regionally variable (good in tumor core, bad at tumor-stroma interface), Geary's C of the errors flags these boundary regions where CCI inference is least reliable.

7. Neighborhood Enrichment (NE Score)

Categorical Statistic | Validates Downstream Biology

Question

"Do specific cell types cluster near each other more than expected by chance?"

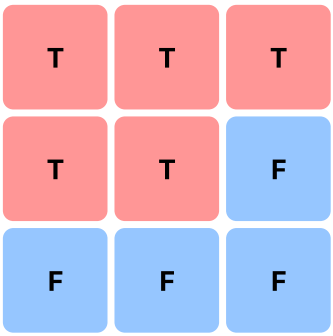
NEIGHBORHOOD ENRICHMENT SCORE

$$NE(a,b) = \frac{\text{observed}(a,b) - \mathbb{E}_{\text{perm}}[\text{count}(a,b)]}{\text{Std}_{\text{perm}}[\text{count}(a,b)]}$$

Term	Meaning
$\text{observed}(a,b)$	Number of edges where type a neighbors type b
$\mathbb{E}_{\text{perm}}$	Expected count under random label permutations
Std_{perm}	Standard deviation under permutations

Worked Example

9 cells, 5 Tumor (T) and 4 Fibroblast (F):



COMPUTATION

Observed T-T neighbor edges: **5**

Expected under random permutation:

$$\mathbb{E}[\text{T-T}] \approx E \times \frac{\binom{n_T}{2}}{\binom{N}{2}} = 12 \times \frac{10}{36} = 3.33$$

Permutation std ≈ 1.2

$$NE(T,T) = \frac{5 - 3.33}{1.2} = \mathbf{1.39}$$

NE Score	Meaning
> 0 (positive)	Cell types cluster together Enriched
≈ 0	Random mixing Neutral
< 0 (negative)	Cell types avoid each other Depleted

Role in your framework: NE scores validate whether spatial cell type organization survives imputation. If imputation errors cause cell type misclassification, NE scores change — indicating the imputation has corrupted the spatial biology.

8. Spatial Statistics Comparison

Property	Moran's I	Geary's C	NE Score
Input type	Continuous (expression)	Continuous (expression)	Categorical (cell types)
Measures	Global autocorrelation	Local dissimilarity	Cell type co-localization
Null model	Random permutation	Random permutation	Label permutation
Range	$[-1, +1]$	$[0, +2]$	$(-\infty, +\infty)$
Clustering signal	$I \to +1$	$C \to 0$	$NE \gg 0$
Random signal	$I \to 0$	$C \to 1$	$NE \approx 0$

Your framework use	Estimate ρ_{lr} for bias correction	Detect local error heterogeneity	Validate downstream biology
Component	Regression Calibration	Reliability Scoring	Downstream Validation

9. Cell-Cell Interaction Test Statistic

CCI SCORE FOR LIGAND-RECEPTOR PAIR (L, R)

$$T_{lr} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} x_{il} \cdot x_{jr}$$

where $\mathcal{E}$ is the set of spatially adjacent cell pairs (edges in the spatial graph).

Significance Testing

PERMUTATION TEST

Under the null hypothesis of no spatial interaction:

1. Randomly permute cell labels: $\pi \sim \mathrm{Uniform}(S_N)$
2. Compute null statistic: $T_{lr}^{(\pi)} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} x_{\pi(i),l} \cdot x_{j,r}$
3. Repeat B times
4. $p\text{-value} = \frac{\#\{T_{lr}^{(\pi)} \geq T_{lr}\}+1}{B+1}$

Multiple testing correction via Benjamini-Hochberg: reject if $p_{\text{adj}} < 0.05$.

10. Bias Decomposition (The Core Insight)

WHY IMPUTATION INFLATES CCI

If the imputed expression is $\hat{\lambda}_{ig} = \lambda_{ig} + e_{ig}$, then:

$$\mathbb{E}[\hat{\lambda}_{il} \cdot \hat{\lambda}_{jr}] = \underbrace{\lambda_{il} \cdot \lambda_{jr}}_{\text{true signal}} + \underbrace{\lambda_{il} \cdot \mathbb{E}[e_{jr}] + \lambda_{jr} \cdot \mathbb{E}[e_{il}]}_{= 0 \text{ if unbiased}} + \underbrace{\mathrm{Cov}(e_{il}, e_{jr})}_{\text{BIAS}}$$

The problem: For graph-based imputation methods (GNN, diffusion, any method using spatial neighbors), $\mathrm{Cov}(e_{il}, e_{jr}) > 0$ when cells i and j are neighbors. The imputation algorithm shares information across spatial edges — the exact same edges used to compute CCI. This creates a *systematic positive bias* in the CCI statistic.

CCI BIAS DECOMPOSITION

$$\mathbb{E}[\hat{T}_{lr}] = T_{lr}^{\text{true}} + \underbrace{\frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \mathrm{Cov}(e_{il}, e_{jr})}_{\text{bias}>0 \text{ for graph-based methods}}$$

11. Regression Calibration

Core Contribution Paper Lead

BIAS ESTIMATION VIA MORAN'S I

Estimate spatial error correlation for each gene:

$$\hat{\rho}_g = I_g = \frac{N}{W} \cdot \frac{\sum_{i,j} w_{ij}(\hat{\lambda}_{ig} - \bar{\hat{\lambda}}_g)(\hat{\lambda}_{jg} - \bar{\hat{\lambda}}_g)}{\sum_i (\hat{\lambda}_{ig} - \bar{\hat{\lambda}}_g)^2}$$

Cross-gene spatial error correlation:

$$\hat{\rho}_{lr} = \sqrt{|\hat{\rho}_l| \cdot |\hat{\rho}_r|}$$

Bias estimate:

$$\widehat{\text{bias}}_{lr} = \hat{\rho}_{lr} \cdot \sqrt{\mathrm{MSE}(l) \cdot \mathrm{MSE}(r)}$$

CORRECTED CCI STATISTIC

$$\hat{T}_{lr}^{\text{corrected}} = \hat{T}_{lr}^{\text{naive}} - \widehat{\text{bias}}_{lr}$$

Corrected variance:

$$\mathrm{Var}(\hat{T}_{lr}^{\text{corrected}}) \approx \mathrm{Var}(\hat{T}_{lr}^{\text{naive}}) + \frac{\mathrm{MSE}(l) \cdot \mathrm{MSE}(r)}{|\mathcal{E}|}$$

12. Corrected Inference Procedure

COMPLETE ALGORITHM

1. Compute naive CCI: $\hat{T}_{lr}$ on imputed data
2. Estimate per-gene MSE: $\widehat{\mathrm{MSE}}(g)$ via cross-validation on measured genes
3. Compute Moran's I for ligand and receptor: $\hat{\rho}_l, \hat{\rho}_r$
4. Estimate bias: $\widehat{\mathrm{bias}}_{lr} = \sqrt{|\hat{\rho}_l||\hat{\rho}_r|} \cdot \sqrt{\widehat{\mathrm{MSE}}(l) \cdot \widehat{\mathrm{MSE}}(r)}$
5. Correct: $\hat{T}_{lr}^{\text{corr}} = \hat{T}_{lr} - \widehat{\mathrm{bias}}_{lr}$
6. Permutation test on corrected statistic (subtract same bias from null)
7. BH correction for multiple testing

13. Error Propagation: Cell Type Classification

MISCLASSIFICATION BOUND

For cell i of true type t, classified via nearest centroid on reconstructed expression:

$$P(\text{misclassify cell } i) \leq \exp\left(-\frac{\Delta_t^2}{8 \cdot \sigma_{\text{recon}}^2}\right)$$

where:

$$\Delta_t = \min_{t' \neq t} \|\boldsymbol{\mu}^{(t)} - \boldsymbol{\mu}^{(t')}\|_2 \quad \text{(minimum inter-type distance)}$$

$$\sigma_{\text{recon}}^2 = \sum_{g=1}^{G} \mathrm{MSE}(g) \quad \text{(total reconstruction variance)}$$

Prediction: Cell types with small Δ_t (e.g., CD4 naive vs CD4 memory T cells) fail first as panel size m decreases. The critical panel size for type t:

$$m_t^* \approx \frac{8\,K\,\lambda_{\max}}{\Delta_t^2}$$

14. Error Propagation: CCI False Positive Rate

VARIANCE INFLATION OF CCI STATISTIC

$$\mathrm{Var}(\hat{T}_{lr}) \geq \frac{1}{|\mathcal{E}|}\left[\bar{\lambda}_r^2 \cdot \mathrm{MSE}(l) + \bar{\lambda}_l^2 \cdot \mathrm{MSE}(r) + \mathrm{MSE}(l) \cdot \mathrm{MSE}(r)\right]$$

Scenario	Variance Scaling	Reliability
Both l, r measured ($\in S$)	Poisson noise only	High
One imputed, one measured	$\propto \mathrm{MSE}(\text{imputed gene})$	Moderate
Both l, r imputed ($\notin S$)	$\propto \mathrm{MSE}(l) \times \mathrm{MSE}(r)$	Low (quadratic)

Quadratic degradation: CCI involving two imputed genes is quadratically less reliable than CCI with one measured gene. This is the strongest argument for panel-aware CCI analysis.

15. Error Propagation: Spatial Domain Detection

SPATIAL DOMAIN STABILITY

Spatial domain detection typically uses Leiden/Louvain clustering on a spatial + expression graph. The clustering output depends on edge weights:

$$w_{ij}^{\mathrm{eff}} = \mathrm{sim}(\hat{\boldsymbol{\lambda}}_i, \hat{\boldsymbol{\lambda}}_j) \cdot \mathrm{spatial_weight}(i,j)$$

Reconstruction error perturbs the similarity:

$$\mathrm{sim}(\hat{\boldsymbol{\lambda}}_i, \hat{\boldsymbol{\lambda}}_j) = \mathrm{sim}(\boldsymbol{\lambda}_i, \boldsymbol{\lambda}_j) + O\left(\frac{\sigma_{\mathrm{recon}}}{\Delta_{\min}}\right)$$

Domains merge or split when $\sigma_{\text{recon}}/\Delta_{\min} \gtrsim 1$.

16. Panel Perturbation Stability Analysis

NOT a Bootstrap | Stability Measure

PROCEDURE

1. From panel S (m genes), create B sub-panels $S_1^*, \ldots, S_B^*$ by dropping fraction α of genes
2. For each S_b^*: re-impute $\hat{\mathbf{\Lambda}}_b$, re-compute CCI $\hat{T}_{lr}^{(b)}$
3. Compute per-pair stability metrics

STABILITY METRICS

Coefficient of Variation

$$\text{CV}_{lr} = \frac{\text{Std}_b[\hat{T}_{lr}^{(b)}]}{|\text{Mean}_b[\hat{T}_{lr}^{(b)}]| + \epsilon}$$

Stability Score

$$\text{Stability}_{lr} = 1 - \text{clip}(\text{CV}_{lr}, 0, 1)$$

Robustness

$$\text{Robustness}_{lr} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}[\hat{T}_{lr}^{(b)} > 0]$$

Stability	Meaning	Action
≈ 1.0	Robust to panel composition	Trust
≈ 0.5	Fragile — depends on which genes are included	Flag
≈ 0.0	Unstable — conclusion is noise	Discard

17. Calibration Analysis

CALIBRATION PLOT CONSTRUCTION

1. Bin significant CCI pairs by predicted reliability (stability score)
2. In each bin b, compute **False Discovery Proportion**:

$$\mathrm{FDP}(b) = 1 - \frac{\backslash\# \text{ true positives in bin } b}{\backslash\# \text{ total in bin } b}$$

Perfect calibration: FDP decreases monotonically as reliability increases.

Target: Spearman correlation between predicted reliability and actual error ≥ 0.7. FDP in highest reliability bin < 0.1. FDP in lowest reliability bin > 0.5.

18. Reliability Certificate

PER-CONCLUSION RELIABILITY OUTPUT

Given panel S, for each biological conclusion c:

$$R(c \mid S) = f\left(\frac{K}{m},\ \{R_g^2\}_{g \in \mathrm{genes}(c)},\ \{\mathrm{MSE}(g)\}_{g \in \mathrm{genes}(c)},\ \mathrm{Stability}(c)\right)$$

Example Output

Conclusion	Reliability	Status
Cell types {Tumor, Fibroblast, Macrophage}: identifiable	0.94	Reliable
Cell type {CD4+ T naive}: identifiable	0.31	Unreliable
CCI: SPP1–CD44 interaction	0.87	Reliable
CCI: CXCL12–CXCR4 interaction	0.23	Unreliable (both imputed)
Spatial domain: 5 domains at m=300	0.78	Moderate

19. Interactive Error Simulator

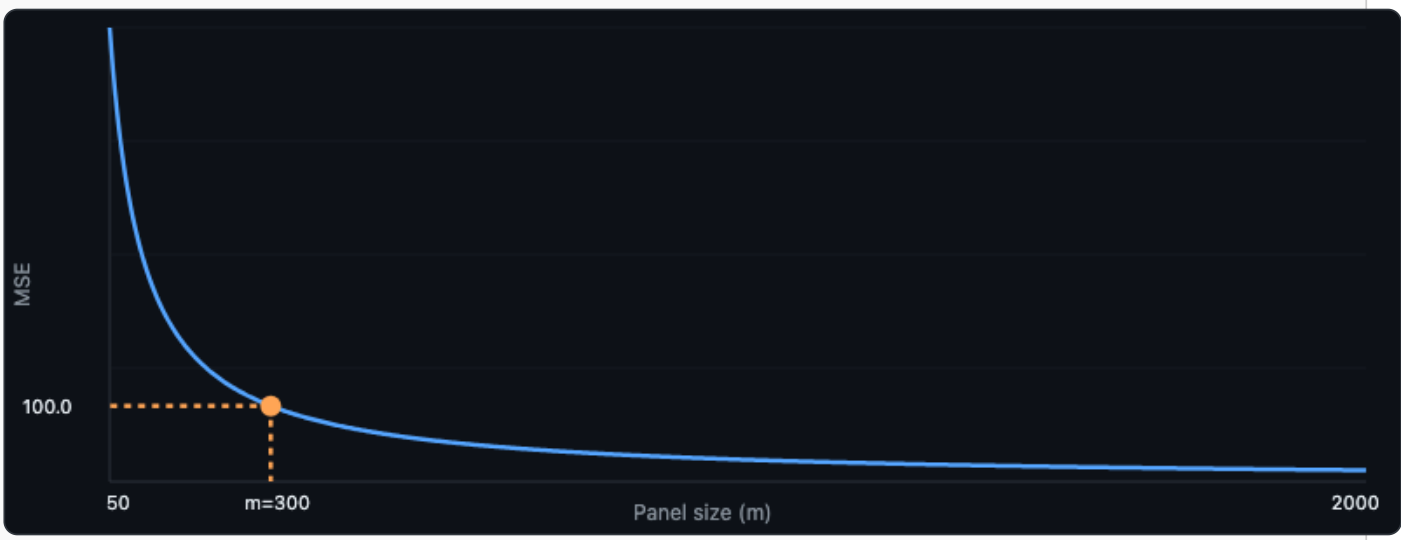
EXPLORE: MSE VS PANEL SIZE AND RANK

$$\text{MSE} \approx C \cdot \frac{K \cdot \lambda_{\max}}{m}$$

Gene programs (K): **30**

Panel size (m): **300**

Max expression ($\lambda_{\max}$): **500**



MSE ≈ **100.00** | K/m = **0.100** | Marginal

EXPLORE: CCI BIAS VS SPATIAL AUTOCORRELATION

$$\text{bias}_{lr} = \rho_{lr} \cdot \sqrt{\text{MSE}(l) \cdot \text{MSE}(r)}$$

Moran's I of ligand (ρ_l): **0.50**

Moran's I of receptor (ρ_r): **0.50**

MSE(ligand): **5.0**

MSE(receptor): **5.0**

ρ_{lr} = **0.500** | bias = **2.500** | HIGH bias — CCI unreliable

20. Go / No-Go Decision Criteria

Criterion	Threshold	Metric
CCI inflation rate	$> 20\%$ across ≥ 2 methods	$\frac{n_{\text{sig,imputed}} - n_{\text{sig,truth}}}{n_{\text{sig,truth}}}$
False positive reduction	$> 50\%$ removed by calibration	$\frac{n_{\text{FP,naive}} - n_{\text{FP,corrected}}}{n_{\text{FP,naive}}}$
Reliability-error correlation	Spearman $r > 0.5$	$r_s(\text{Stability}, \lvert\hat{T} - T_{\text{true}}\rvert)$
Calibration	Monotonic FDP decrease	FDP per reliability bin

DECISION MATRIX

Result	Verdict	Target Venue
All 4 criteria met, large effect	STRONG GO	Genome Biology / Nature Methods
3 of 4 criteria met	GO	Bioinformatics / Biostatistics
1–2 criteria met	WEAK	Bioinformatics (with caveats)
0 criteria met	NO-GO	Pivot project direction

21. Complete Notation Reference

Symbol	Type	Definition
N	Scalar	Number of cells

G	Scalar	Total genes in genome
m	Scalar	Number of measured genes ($\|S\|$)
K	Scalar	Number of latent gene programs (rank)
S	Set	Measured gene panel $\subset \{1, \ldots, G\}$
$\mathbf{s}_i$	$\mathbb{R}^2$	Spatial coordinates of cell i
y_{ig}	$\mathbb{Z}_+$	Observed transcript count
λ_{ig}	$\mathbb{R}_+$	True expression rate
$\hat{\lambda}_{ig}$	$\mathbb{R}_+$	Imputed expression rate
e_{ig}	$\mathbb{R}$	Imputation error ($\hat{\lambda}_{ig} - \lambda_{ig}$)
δ_g	$(0, 1]$	Detection efficiency for gene g
$\mathbf{U}$	$\mathbb{R}^{N \times K}$	Cell loading matrix
$\mathbf{V}$	$\mathbb{R}^{G \times K}$	Gene program matrix
$\mathbf{v}_g$	$\mathbb{R}^K$	Gene g's program membership vector
R_g^2	$[0, 1]$	Gene g's predictability from panel S
w_{ij}	$\{0, 1\}$	Spatial adjacency (1 if neighbors)
$\mathcal{E}$	Set	Set of spatial edges (neighbor pairs)
T_{lr}	$\mathbb{R}$	CCI test statistic for pair (l, r)
I_g	$[-1, 1]$	Moran's I for gene g
C_g	$[0, 2]$	Geary's C for gene g
ρ_{lr}	$[0, 1]$	Cross-gene spatial error correlation
Δ_t	$\mathbb{R}_+$	Minimum inter-cell-type distance

22. Complete Model Summary

THE ENTIRE FRAMEWORK IN ONE BOX

Observation:	$y_{ig} \sim \text{Poisson}(\delta_g \lambda_{ig}), \quad g \in S,\ \lvert S\rvert = m$
Structure:	$\log \lambda_{ig} = \mu_g + \sum_{k=1}^{K} u_{ik}\, v_{gk} + \epsilon_{ig}$
Recovery MSE:	$\frac{1}{NG} \lVert \hat{\mathbf{\Lambda}} - \mathbf{\Lambda} \rVert_F^2 \le C\, \frac{K \lambda_{\max}}{m}$
Per-gene error:	$\text{MSE}(g) = \frac{1}{\delta_g \bar{\lambda}_g} + \lVert \mathbf{v}_g \rVert^2 (1 - R_g^2)$
CCI bias:	$\mathbb{E}[\hat{T}_{lr}] = T_{lr}^{\text{true}} + \rho_{lr} \sqrt{\text{MSE}(l)\, \text{MSE}(r)}$
Corrected CCI:	$\hat{T}_{lr}^{\text{corr}} = \hat{T}_{lr} - \hat{\rho}_{lr} \sqrt{\widehat{\text{MSE}}(l)\, \widehat{\text{MSE}}(r)}$
Cell typing:	$P(\text{misclassify}) \le \exp\left(-\frac{\Delta_t^2}{8 \sum_g \text{MSE}(g)} \right)$
CCI variance:	$\text{Var}(\hat{T}_{lr}) \propto \text{MSE}(l) \cdot \text{MSE}(r) \ \text{ if both imputed}$

Three governing quantities:

K/m	Rank over panel size — can you capture the latent structure?
R_g^2	Per-gene predictability — is this gene in the measured subspace?
Δ_t **/ effect size**	Biological signal strength — can biology survive the noise?

Dashboard generated for: *Measurement Error Framework for Spatial Omics Inference*

Framework version 1.0 • All formulations are paper-ready